

LEVEL-VI

SINGLE ANSWER QUESTIONS

- Let $f: R \rightarrow R$ defined by $f(x) = \frac{e^{x^2} - e^{-x^2}}{e^{x^2} + e^{-x^2}}$ then $f(x)$ is
 a) one-one but not onto
 b) neither one-one nor onto
 c) many-one but onto
 d) one-one but not onto
- If the graph of a function $f(x)$ is symmetrical about the line $x=a$ then
 a) $f(a+x) = f(a-x)$
 b) $f(a+x) = f(x-a)$
 c) $f(x) = f(-x)$
 d) $f(x) = -f(-x)$
- If $f(x) = f(2a-x)$ then the graph of $f(x)$ is symmetric about the line
 a) $y=a$
 b) $x=2a$
 c) $x=a$
 d) $x=-a$
- There are exactly two linear functions which map from $[-1,1]$ onto $[0,2]$ they are
 a) $y=x+1, y=x-1$
 b) $y=x+1, y=-x+1$
 c) $y=2x-1, y=x-4$
 d) $y=x, y=3x$
- $f: (-\infty, -1] \rightarrow (0, e^5]$ defined by $f(x) = e^{x^3-3x+2}$ is
 a) one-one and into
 b) one-one and onto
 c) many-one and into
 d) many-one and onto
- Let $f(x) = x + 2|x+1| + 2|x-1|$. If $f(x) = k$ has exactly one real solution, then k equals
 a) 3
 b) 0
 c) 1
 d) 2
- Let $f(1) = 1$ and $f(n) = 2 \sum_{r=1}^{n-1} f(r)$ then $\sum_{n=1}^m f(n) =$
 a) $3^m - 1$
 b) 3^m
 c) 3^{m-1}
 d) 3^{m-2}
- The domain of $f(x) = \log_{10} \{1 - \log_{10}(x^2 - 5x + 16)\}$ is
 a) $(2,3)$
 b) $(0,\infty)$
 c) $[1,3]$
 d) $[2,3]$

- If $[.]$ denotes G.I.F then the domain of $f(x) = \cos^{-1}(x + [x])$ is
 a) $(0,1)$
 b) $[0,1)$
 c) $[0,1]$
 d) $[-1,1]$
- The domain of $f(x) = \cot^{-1} \left(\frac{x}{\sqrt{x^2 - [x^2]}} \right)$ (where $[.]$ is G.I.F)
 a) R
 b) $R - \{0\}$
 c) $R - \{\pm\sqrt{n}, n \in Z \text{ and } n \geq 0\}$
 d) None
- The domain of $f(x) = \frac{1}{\ln[\cos^{-1} x]}$ (where $[.]$ G.I.F) is
 a) $[0,1]$
 b) $[-1, \cos 2]$
 c) $[-1, \cos 3) \cup (\cos 3, \cos 4)$
 d) $[-1, \cos 3) \cup (\cos 3, \cos 2)$
- If $f(x) = \cos^{-1}(x - x^2) + \sqrt{\left(1 - \frac{1}{|x|}\right)} + \frac{1}{x^2 - 1}$ then domain of $f(x)$ (where $[.]$ G.I.F) is
 a) $\left(\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$
 b) $\left(-\sqrt{2}, \frac{1-\sqrt{5}}{2}\right)$
 c) $\left[\sqrt{2}, \frac{1+\sqrt{5}}{2}\right]$
 d) $\left[\sqrt{7}, \frac{\sqrt{7}}{2}\right]$
- The range of $f(x) = \sin^{-1}(\sqrt{x^2 + x + 1})$ is
 a) $\left[0, \frac{\pi}{2}\right]$
 b) $\left[0, \frac{\pi}{3}\right]$
 c) $\left[\frac{\pi}{3}, \frac{\pi}{2}\right]$
 d) $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$
- The function $f(x) = |px - q| + r|x|, x \in (-\infty, \infty)$ where $p > 0, q > 0, r > 0$ assumes its minimum value only at one point if
 a) $p \neq q$
 b) $r \neq q$
 c) $r \neq p$
 d) $p = q = r$

15. Let f, g, h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$; $g(x) = x.e^{x^2} + e^{-x^2}$ and $h(x) = x^2.e^{x^2} + e^{-x^2}$. If a, b, c denote respectively the absolute max. values of f, g, h on $[0, 1]$ then
- a) $a = b$ and $c \neq b$ b) $a = c$ and $a \neq b$
 c) $a \neq b$ and $c \neq b$ d) $a = b = c$

16. The range of $f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$ is

a) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ b) $[0, \pi]$ c) $\left[0, \frac{\pi}{2}\right)$ d) $(0, \pi)$

17. Let 'f' be an injective mapping with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$ such that exactly one of the following statements is correct and the remaining are false
 $f(x) = 1; f(y) \neq 1; f(z) \neq 2$ then $f^{-1}(1) =$

[IIT 1982]

a) x b) y c) z d) None

18. If $f(x) = -1 + |x - 2|; 0 \leq x \leq 4$
 $g(x) = 2 - |x|; -1 \leq x \leq 3$ then $(f \circ g)(x) =$
 $-1 - x; -1 \leq x \leq 0$ $-1 + x; 0 \leq x \leq 1$
 a) $x - 1; 0 < x \leq 2$ b) $x - 1; 1 < x \leq 2$
 c) $-1 - x; -1 \leq x \leq 2$ d) does not exist

19. If $f: [1, \infty) \rightarrow [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$ then $f^{-1}(x) =$

a) $\frac{x \pm \sqrt{x^2 - 3}}{4}$ b) $\frac{x \pm \sqrt{x^2 - 4}}{2}$
 c) $\frac{x + \sqrt{x^2 - 4}}{2}$ d) $\frac{x - \sqrt{x^2 - 4}}{2}$

20. The period of the real-valued function satisfying

$$f(x) + f(x+4) = f(x+2) + f(x+6) \text{ is}$$

a) 10 b) 8 c) 12 d) 6

21. Let $f(x)$ be a periodic function with period 3 and $f\left(\frac{-2}{3}\right) = 7$ and $g(x) =$

$$\int_0^x f(t+n) dt \text{ where } n = 3K, K \in \mathbb{N} \text{ then}$$

$$g^1\left(\frac{7}{3}\right) =$$

a) $\frac{-2}{3}$ b) 7 c) -7 d) $\frac{7}{3}$

22. If $f(x)$ is an even function and satisfies the relation $x^2 \cdot f(x) - 2 \cdot f\left(\frac{1}{x}\right) = g(x)$ where $g(x)$ is an odd function, then the value of $f(5)$ is

a) 0 b) $\frac{37}{55}$ c) 4 d) 5

23. Consider a real valued function $f(x)$ satisfying $2f(xy) = (f(x))^y + (f(y))^x$
 $\forall x, y \in \mathbb{R}$ and $f(1) = a$ where $a \neq 1$ then

$$(a-1) \sum_{i=1}^n f(i) =$$

a) $a^{n+1} + a$ b) a^{n+1} c) $a^{n+1} - a$ d) $a^{n+2} - a$

24. If 'p' and 'q' are +ve integers, f is a function defined for +ve numbers and attains only +ve values such that

$$f(x, f(y)) = x^p y^q \text{ then } p^2 =$$

a) 2q b) q c) 3q d) 4q

25. If the function 'f' satisfies the relation

$$f(x+y) + f(x-y)$$

$$= 2f(x)f(y) \quad \forall x, y \in \mathbb{R} \text{ and } f(0) \neq 0$$

then $f(x)$ is an

- a) even function
 b) odd function
 c) Neither even nor odd
 d) can not decide

26. A real valued function $f(x)$ satisfies the function

$$f(x-y) = f(x) \cdot f(y) - f(a-x)f(a+y)$$

where 'a' is a given constant and $f(0) = 1$,

FUNCTIONS

JEE ADVANCED - VOL - III

then the graph of the function is symmetrical about [IIT 2005]

- a) point $(2a, 0)$ b) point $(a, 0)$
c) line $x = 2a$ d) line $x = a$

27. The function 'f' satisfies the functional

equation $3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$

for all real $x \neq 1$, then the value of $f(7)$ is

- a) 8 b) 4 c) -8 d) 11

28. Let 'f' be a real valued function defined for all $x \in R$ such that for some fixed $a > 0$,

$f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ for all 'x'

then the period of $f(x)$ is

- a) $\frac{a}{4}$ b) $\frac{a}{3}$ c) $2a$ d) None

29. Let $f(x, y)$ be a periodic function satisfying

$f(x, y) = f(2x+2y, 2y-2x) \forall x, y \in R$

defined $g(x) = f(2^x, 0)$ then the period of

$g(x)$ is

- a) 4 b) 6 c) 8 d) 12

30. If $f(x) = \min(x^2, x, \operatorname{sgn}(x^2 + 4x + 5))$

then the value of $f(2)$ is equal to

- a) 2 b) 0 c) 1 d) 4

31. If $f(x) = \sin^2 x + \sin^2\left(\frac{\pi}{3} + x\right)$

$+\cos(x) \cos\left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$

then the graph of $y = g(f(x))$ is

- a) a circle b) a straight line
c) a parabola d) a pair of straight lines

32. Let $f_1(n) = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ then

$f_1(1) + f_1(2) + f_1(3) + \dots + f_1(n) =$

a) $n \cdot f_1(n) - 1$ b) $(n+1)f_1(n) + n$

c) $(n+1)f_1(n) - n$ d) $n \cdot f_1(n) + n$

MULTIPLE ANSWER QUESTIONS

33. If $f(x) = x|x|$; $0 \leq x < 1$

$= 2x$; $x \geq 1$ then its

a) even extension is b) odd extension is

x^2 ; $-1 \leq x \leq 0$ x^2 ; $-1 \leq x \leq 0$

$-2x$; $-\infty < x \leq -1$ $2x$; $-\infty < x \leq -1$

c) even extension d) does not exist

x^2 ; $0 \leq x \leq 1$

$-2x$; $1 < x < \infty$

34. If $f(x) = \sqrt{3|x| - x - 2}$ and $g(x) = \sin x$

then the domain of $(f \circ g)(x) =$

a) $\left\{2m\pi + \frac{\pi}{2}\right\}; m \in Z$

b) $\left[2m\pi + \frac{7\pi}{6}, 2m\pi + \frac{11\pi}{6}\right]; m \in Z$

c) $\left\{2m\pi + \frac{\pi}{3}\right\}; m \in Z$ d) ϕ

35. The domain of $f(x) = \frac{1}{\sqrt{[|x| - 1] - 5}}$

(where $[.]$ G.I.F) is

a) $[-7, 7]$ b) $(-\infty, 7]$

c) $(-\infty, -7]$ d) $[7, \infty)$

36. For the function $f(x)$ satisfying

$2.f(\sin x) + f(\cos x) = x, \forall x \in R$

a) Domain is $[0, 1]$ b) range is $\left[\frac{-2\pi}{3}, \frac{\pi}{3}\right]$

c) Domain is $[-1, 1]$ d) range is $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

37. Let $f(x) = \sin x$, $g(x) = \ln |x|$ If the ranges of f and g are R_1 and R_2 respectively then [IIT 1994]

- a) $R_1 = \{U : -1 \leq U \leq 1\}$
 b) $R_2 = \{V : -\infty < V \leq 0\}$
 c) $R_1 = \{U : 0 \leq U \leq 1\}$ d) None

38. The polynomial $p(x)$ is such that for any polynomial $q(x)$ we have

$p(q(x)) = q(p(x))$ then $p(x)$ is

- a) even b) odd
 c) of even degree d) of odd degree

39. Let $f(x) = \max\{1 + \sin x, 1 - \cos x\}$;
 $x \in [0, 2\pi]$ and $g(x) = \max\{1, |x - 1|\}$;
 $x \in R$ then

- a) $g(f(0)) = 1$ b) $g(f(1)) = 1$
 c) $f(g(1)) = 1$ d) $f(g(0)) = \sin 1$

40. Let $f(x) = \sin x + \cos\left(\left(\sqrt{4 - a^2}\right)x\right)$. Then the integral values of 'a' for which $f(x)$ is a periodic function, are given by

- a) 1 b) 2 c) -2 d) 0

41. If a polynomial of degree 'n' satisfies

$f(x) = f^1(x) \cdot f^{11}(x) \forall n \in R$ then $f(x)$ is

- a) an onto function b) an into function
 c) no such function is possible
 d) even function

42. Let $f(x) = \ln(2x - x^2) + \sin\left(\frac{\pi x}{2}\right)$ then

- a) graph of 'f' is symmetrical about the line $x=1$
 b) graph of 'f' is symmetrical about the line $x=2$
 c) max. value of 'f' is 1
 d) min. value of 'f' does not exist

PARAGRAPH QUESTIONS

- (P) Let $F(x) = f(x) + g(x)$;

$$G(x) = f(x) - g(x) \text{ and } H(x) = \frac{f(x)}{g(x)}$$

where $f(x) = 1 - 2\sin^2 x$ and

$$g(x) = \cos(2x) \forall f: R \rightarrow [-1, 1] \text{ and}$$

$$g: R \rightarrow [-1, 1]$$

Now answer the following:

43. Domain and range of $H(x)$ are

- a) R and $\{1\}$ b) R and $\{0, 1\}$
 c) $R - \left\{(2n+1)\frac{\pi}{4}\right\}$ and $\{1\}$ d) None
 $n \in Z$

44. If $F: R \rightarrow [-2, 2]$ then $F(x)$ is

- a) one-one b) onto c) into d) None

45. Which of the following is correct

a) periods of $f(x)$, $g(x)$ and $F(x)$ makes A.P with the common difference $\frac{\pi}{3}$

b) period of $f(x)$, $g(x)$ and $F(x)$ are same and is equal to 2π

c) sum of the periods of $f(x)$, $g(x)$ and $F(x)$ is 3π

d) sum of the periods of $f(x)$, $g(x)$ and $F(x)$ is 6π

46. Which of the following is correct

- a) the domain of $G(x)$ and $H(x)$ are same
 b) the range of $G(x)$ and $H(x)$ are same
 c) the union of the domain of $G(x)$ and $H(x)$ are all real numbers
 d) None

47. If the solutions of $F(x) - G(x) = 0$ are

$x_1, x_2, x_3, \dots, x_n$ where $x \in [0, 5\pi]$ then

a) $x_1, x_2, x_3, \dots, x_n$ are in A.P with

common difference $\frac{\pi}{4}$

b) the no. of solutions of

$$F(x) - G(x) = 0 \text{ is } 10 \forall x \in [0, 5\pi]$$

c) the sum of all solutions of

$$F(x) - G(x) = 0 \text{ is } \forall x \in [0, 5\pi] \text{ is } 25\pi$$

d) b, c are true

- (P) Let $f: N \rightarrow N$ be a function defined by $f(x)$ = the biggest +ve integer obtained by reshuffling the digits of 'x'. For example $f(296) = 962$. Now answer the following.

48. f is

- a) one-one, onto b) one-one and into
c) many-one and onto d) many-one and into

49. The biggest +ve integer which divides

$$f(n) - n, \forall n \in N \text{ is}$$

- a) 3 b) 9 c) 18 d) 27

50. The range of f is

- a) N
b) set of +ve integers whose digits are non-increasing from left to right
c) set of +ve whose digits are non-decreasing from left to right
d) None

- (P) A function 'f' from a set X to Y is called onto if every $y \in Y, \exists x \in X$ such that $f(x) = y$. Unless the contrary is specified, a real function is onto if it takes all real values. Otherwise, it is called on into function. Thus, if X and Y are finite sets, then 'f' can not be onto if Y contains more elements than 'X'. Now answer the following.

51. The polynomial function

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$$

where $a_0 \neq 0$ is onto, for

- a) all positive integers n
b) all even +ve integers 'n'
c) all odd +ve integers 'n'
d) no +ve integer

52. The function $f(x) = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ is onto, if

- a) $0 < c < 2$ b) $0 < c < 4$
c) $\frac{-1}{2} < c < \frac{1}{2}$ d) $0 < c < 1$

53. Which of the following is not true

- a) A one-one function from the set $\{a, b, c\}$ to $\{\alpha, \beta, \gamma\}$ is onto also
b) An onto function from an infinite set to a finite set can not be one-one
c) An onto function is always invertible
d) the functions $\tan x$ and $\cot x$ are onto

(P) Let $f: R \rightarrow R$ be a continuous function

$$\text{such that } f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{4}\right) = x^2.$$

Now answer the following

54. $f(3) =$

- a) $f(0)$ b) $4 + f(0)$ c) $9 + f(0)$ d) $16 + f(0)$

55. The equation $f(x) - x - f(0) = 0$ have exactly

- a) no solution b) one solution
c) two solution d) infinite solutions

56. $f^1(0) =$

- a) 0 b) 1 c) $f(0)$ d) $-f(0)$

(P) If $(f(x))^2 f\left(\frac{1-x}{1+x}\right) = 64x, x \neq 0$ then

57. $f(x) =$

- a) $4x^{\frac{2}{3}} \left(\frac{1+x}{1-x}\right)^{\frac{1}{3}}$ b) $x^{\frac{1}{3}} \left(\frac{1-x}{1+x}\right)^{\frac{1}{3}}$
c) $x^{\frac{2}{3}} \left(\frac{1-x}{1+x}\right)^{\frac{1}{3}}$ d) $16x^{1/3} \left(\frac{1+x}{1-x}\right)^{1/3}$

58. The domain of $f(x)$ is

- a) $[0, \infty)$ b) $R - \{-1, 1\}$
c) $(-\infty, \infty)$ d) $R - \{0, 1, -1\}$

59. The value of $f\left(\frac{9}{7}\right)$ is

- a) $8\left(\frac{7}{9}\right)^{2/3}$ b) $4\left(\frac{9}{7}\right)^{1/3}$
c) $-8\left(\frac{9}{7}\right)^{2/3}$ d) $-4\left(\frac{9}{7}\right)^{1/3}$

- (P) Based upon each paragraph, three multiple choice questions have to be answered. Each question has four choices a,b,c and d, out of which only one is correct.

Passage: Consider the functions

$$f(x) = \begin{cases} x+1, & x \leq 1 \\ 2x+1, & 1 < x \leq 2 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x^2, & -1 \leq x < 2 \\ x+2, & 2 \leq x \leq 3 \end{cases}$$

60. The domain of the function $f(g(x))$ is
 a) $[0, \sqrt{2}]$ b) $[-1, 2]$
 c) $[-1, \sqrt{2}]$ d) None of these
61. The range of the function $f(g(x))$ is
 a) $[1, 5]$ b) $[2, 3]$
 c) $[1, 2] \cup (3, 5]$ d) none of these
62. The number of roots of the equation $f(g(x)) = 2$ is
 a) 1 b) 2
 c) 4 d) None of these
- (P) Let $f(x) = \sin x - x \cos x, \forall x \in R$. Now answer the following.
63. The least +ve value of 'x' for which $f(x)=0$ lies in the quadrant
 a) Q_1 B) Q_2 C) Q_3 D) Q_4
64. The set of all the values of $x \in (0, 2\pi)$ for which $f(x) > 0$ is
 a) $(0, \pi)$ b) $(\pi, 2\pi)$ c) $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ d) None
65. If α is the least +ve value for which $\tan \alpha = \alpha$ then the area bounded by $y = f(x)$, X -axis, $x = 0$ and $x = 2\pi$ is
 a) 4 b) $4(1 - \cos \alpha)$
 c) $4(1 + \cos \alpha)$ d) $4 - 2(2 + \alpha^2) \cos \alpha$

MATRIX-MATCHING QUESTIONS

66. Match the following:

COLUMN-I

a) $f: R \rightarrow \left[\frac{\pi}{4}, \pi\right)$ and

$f(x) = \cot^{-1}(2x - x^2 - 2)$ then $f(x)$ is

b) $f: R \rightarrow R$ and $f(x) = e^{ax} \sin(bx)$ where $a, b \in R^+$ then $f(x)$ is

c) $f: R^+ \rightarrow [2, \infty)$ and $f(x) = 2 + 3x^2$ then $f(x)$ is

d) $f: X \rightarrow X$ and $f(f(x)) = x \forall x \in X$ then $f(x)$ is

COLUMN-II

p) one-one

q) into

r) many-one

s) onto

67. Match the following:

LIST-I

a) the integral value of $x \in (-\pi, \pi)$ satisfying

the equation $|x^2 - 1 + \cos x| = |x^2 - 1| + |\cos x|$ can be

b) the number of solutions of $[x]^2 = x + 2\{x\}$ is (where $[.]$ is G.I.F, $\{.\}$ is fractional part functions)

c) If $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x$ then

$[f(x)]$ is

d) An allowable value of $f(x) = \sqrt{\ln(\cos(\sin x))}$

LIST-II

p) 0

q) 1

r) 2

s) 4

t) -1

68. Match the following equations with their number of roots

LIST-ILIST-II

- a) $x^2 \tan x = 1; x \in [0, 2\pi]$ p) 5
 b) $2^{\cos x} = |\sin x|; x \in [0, 2\pi]$ q) 2
 c) If $f(x)$ is a polynomial of degree 5 with real coefficients such that $f(|x|) = 0$ has 8 real roots then the number of roots of $f(x) = 0$
 d) $7^{|x|} (|15 - |x||) = 1$ s) 4

69. Match the following:

COLUMN-I

- a) If $f(x) = x + 1$, when $-1 \leq x < 0$
 $= x^2 - 1$; when $x \geq 0$ then $(f \circ f)(x)$ for $-1 \leq x < 0$ is
 b) If $f\left(\frac{2 \tan x}{1 + \tan^2 x}\right)$
 $= \frac{(\cos(2x) + 1)(\sec^2 x + 2 \tan x)}{2}$ then $f(x)$ is
 c) If $f(x + y + 1) = (\sqrt{f(x)} + \sqrt{f(y)})^2 \forall x,$
 $y \in R$ and $f(0) = 1$ then $f(x)$ is
 d) If $4 < x < 5$ and $f(x) = \left[\frac{x}{4}\right] + 2x + 2$

(Where $[.]$ is G.I.F) then $f^{-1}(x)$ is

COLUMN-II

- p) $\frac{x-3}{2}$
 q) $x^2 + 2x$
 r) $1+x$
 s) $(x+1)^2$

ASSERTION - REASON QUESTIONS

The following questions consist of two statements, one labelled as 'Statement-I' and the other 'Statement-II'. You are to examine these two statements carefully and decide if the Statement-I and the Statement-II are individually true and if so, whether the Statement-II is the correct explanation for the given Statement-I. Select your answer to these items using the codes given below and then select the correct option.

- (A) Both S-I and S-II are individually true and R is the correct explanation of A
 (B) Both S-I and S-II are individually true but S-II is not the correct explanation of S-I
 (C) S-I is true but S-II is false
 (D) S-I is false but S-II is true

70. Consider two functions $f(x) = 1 + e^{\cot^2 x}$ and

$$g(x) = \sqrt{2|\sin x| - 1} + \frac{1 - \cos(2x)}{1 + \sin^4 x}$$

S-I: The solution of $f(x) = g(x)$ is given by

$$x = (2n+1)\frac{\pi}{2}; \forall n \in I$$

S-II: If $f(x) \geq k$ and $g(x) \leq k$ (where $K \in R$) then the solutions of the equation $f(x) = g(x)$ is the solution corresponding to $f(x) = K$

71. S-I: $f: R \rightarrow [-1, 1], f(x) = \cos x$ is not a bijection
 S-II: Every even function is not one-one
 72. S-I: The function $f(x) = |x|$ is not one-one
 S-II: The negative real numbers are not the pre images of any real numbers
 73. S-I: If $f(x) = x^5 - 16x + 2$ then $f(x) = 0$ has only one root in $[-1, 1]$
 S-II: $f(-1)$ and $f(1)$ are of opposite signs

INTEGER QUESTIONS

74. The period of the function satisfying the relation $f(x) + f(x+3) = 0, \forall x \in R$ is

75. If the period of $\frac{\cos(\sin(nx))}{\tan\left(\frac{x}{n}\right)}$; $n \in N$ is 6π

then 'n' is

76. Let $f(x) = x(2-x); 0 \leq x \leq 2$. If the definition of 'f' is extended over the set $R-[0,2]$ by $f(x+1) = f(x)$ then f is a periodic function with period.

77. If $f(x)$ and $g(x)$ are any two real valued functions such that $|f(x) + g(x)| \geq |f(x)| + |g(x)|$ and $g(x) \neq 0$,

$f(x)g(x) \leq 0$ then the value of $\sum_{r=1}^{100} f(r) =$

78. The period of $f(x) = \sin(\sin(\pi x)) + e^{\{3x\}}$ is (where $\{.\}$ is fractional part function).

79. If $f(x) = \operatorname{sgn}(x^2 - 2x + 3)$ then the value of $f(x)$ is

80. Given $y = 2[x] + 3$ and $y = 3[x-2] + 5$ (where $[.]$ denotes the greatest integer function) then the value of $[x+3y]$ is 37λ then λ is

81. If the range of the function

$f(x) = \frac{(1+x+x^2)(1+x^4)}{x^3}$ when $x > 0$ is $[K, \infty)$ then K is

82. If fractional parts of $\frac{1}{x}$ and x^2 for some $x \in (\sqrt{2}, \sqrt{3})$ are equal then the value of $x^4 - \frac{3}{x}$ is

83. A non-zero function $f(x)$ is symmetrical about the line $y = x$ then the value of λ (constant) such that

$f^2(x) = (f^{-1}(x))^2 - \lambda x f(x) f^{-1}(x) + 3x^2 f(x) \forall x \in R^+$ is

84. If $x^2 + 4y^2 = 4$ and range of $f(x, y)$ is $[L, M]$ where $f(x, y) = x^2 + y^2 - xy$ then LM is

85. If the periodic function $f(x)$ satisfies the equation $f(x+1) + f(x-1) = \sqrt{3}f(x)$ $\forall x \in R$ and the period of $f(x)$ is 4λ then $\lambda =$

KEY - LEVEL-VI

SINGLE ANSWER QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 1) B | 2) A | 3) C | 4) B |
| 5) A | 6) B | 7) C | 8) A |
| 9) B | 10) C | 11) B | 12) C |
| 13) C | 14) C | 15) D | 16) C |
| 17) B | 18) A | 19) C | 20) B |
| 21) B | 22) A | 23) C | 24) B |
| 25) A | 26) B | 27) B | 28) C |
| 29) D | 30) C | 31) B | 32) C |

MULTIPLE ANSWER QUESTIONS

- | | | | |
|----------|-------------|----------|-------------|
| 33) A, B | 34) A, B | 35) C, D | 36) B, C |
| 37) A, B | 38) B, D | 39) A, B | 40) B, C, D |
| 41) A | 42) A, C, D | | |

PARAGRAPH QUESTIONS

- | | | | |
|-------|-------|-------|-------|
| 43) C | 44) B | 45) C | 46) C |
| 47) D | 48) D | 49) B | 50) B |
| 51) C | 52) D | 53) C | 54) D |
| 55) C | 56) A | 57) A | 58) B |
| 59) C | 60) C | 61) C | 62) B |
| 63) C | 64) D | 65) D | |

MATRIX-MATCHING QUESTIONS

- 66) A-Q,R; B-R,S; C-P,S; D-P,S
 67) A-Q,T; B-S; C-P,Q,R; D-P
 68) A-Q; B-S; C-P; D-S
 69) A-Q; B-R; C-S; D-P

ASSERTION - REASON QUESTIONS

- 70)C 71)A 72)C 73)B

INTEGER QUESTIONS

- 74)6 75)6 76)2 77)0
 78)2 79)1 80)1 81)6
 82)5 83)3 84)3 85)3

HINTS - LEVEL-VI**SINGLE ANSWER QUESTIONS**

- $e^{x^2} + e^{-x^2} > 0$ and
 $e^{x^2} - e^{-x^2} = 2 \left\{ x^2 + \frac{x^6}{3} + \dots \right\} > 0$
 $f(x) > 0$
 $\Rightarrow$ range $\neq$ co-domain $\Rightarrow$ into function
 Also

$$f^{-1}(x) = \frac{8x}{(e^{x^2} + e^{-x^2})^2} \Rightarrow f^{-1}(x) > 0 \text{ and } f^{-1}(x) < 0$$

if $x > 0$ if $x < 0$

 Not one-one $\rightarrow$ many-one
- Even function is symmetrical about the y-axis i.e. $x=0$
 $\Rightarrow f(x) = f(-x) \Rightarrow f(0+x) = f(0-x)$
 symmetrical about $x=0$
 Similarly symmetrical about
 $x=a \Rightarrow f(a+x) = f(a-x)$
- Put $x=a-x$ on both sides
 $\Rightarrow f(x+a) = f(a-x)$
- $f(-1) = 2$
 $f(1) = 0$ incase of decreasing function

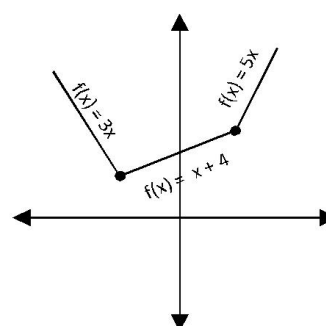
$$-a+b=2$$

$$a+b=0 \Rightarrow b=1, a=-1$$

$$\Rightarrow f(x) = -x+1$$

5. Clearly $f^{-1}(x) = e^{3x^2-3x+2} \cdot 3(x^2-1) \geq 0 \Rightarrow 'f'$ is increasing
 $f(x)$ is one-one
 range of f is $(f(-\infty), f(-1)] =$
 $= (0, e^4) \neq (0, e^5] \Rightarrow 'f'$ is into

6.



$$\text{Let } f(x) = x + |2(x+1)| + 2|x-1|$$

Then

$$f(x) = \begin{cases} x - 2(x+1) - 2(x-1), & x < -1 \\ x - 2(x+1) - 2(x-1), & -1 \leq x \leq 1 \\ x - 2(x+1) + 2(x-1), & x > 1 \end{cases}$$

$$f(x) = \begin{cases} -3x, & x < -1 \\ x + 4, & -1 \leq x \leq 1 \\ 5x, & x > 1 \end{cases}$$

graph of $y = f(x)$ is as shown. Clearly $y=k$ can intersect $y = f(x)$ at exactly one point only if $k=3$.

7. $f(2) = 2 \sum_{r=1}^1 f(1) = 2$
 $f(3) = 2 \sum_{r=1}^2 f(r) = 2[f(1) + f(2)] = 2[1+2] = 6$
 $f(4) = 18 \rightarrow \sum_{n=1}^m f(n) = 3^{m-1}$
8. $1 - \log_{10}(x^2 - 5x + 6) > 0$ and
 $x^2 - 5x + 16 > 0$
 $\log_{10}(x^2 - 5x + 6) < 1$
 $x^2 - 5x + 16 < 10$

$$x \in (2, 3)$$

this is true for all $x \in R$ ($\because \text{Disc} < 0$)

and leading coeff > 0

9. $-1 \leq x + [x] \leq 1$

when $x \in \mathbb{Z}$ i.e. Let $x = k$

$$\Rightarrow -1 \leq 2k \leq 1 \Rightarrow \frac{-1}{2} \leq k \leq \frac{1}{2} \Rightarrow k = 0 \Rightarrow x = 0$$

when $x \notin \mathbb{Z}$ i.e. let $x = k + f \Rightarrow -1 \leq 2k + f \leq 1$

$$\Rightarrow \frac{-1-f}{2} \leq k \leq \frac{1-f}{2} \Rightarrow k = 0 \Rightarrow 0 < x < 1.$$

$$\therefore Df = [0, 1)$$

10. $x^2 - [x^2] > 0$

$$x^2 > [x^2]$$

$$x^2 \in R \text{ and } x^2 \neq 0, 1, 2, \dots$$

$$\therefore \text{Domain is } R - \{\pm\sqrt{n}, n \in \mathbb{Z} \text{ and } n \geq 0\}$$

11. $[\cos^{-1} x] \neq 1 \text{ and } [\cos^{-1} x] > 0$

$$\cos^{-1} x \notin [1, 2) \quad [\cos^{-1} x] = 2$$

$$2 \leq \cos^{-1} x \leq \pi, \cos n \geq x \geq -1, -1 \leq x \leq \cos 2$$

12. Take $-1 \leq x - x^2 \leq 1$ and $1 - \frac{1}{|x|} \geq 0$

$$\text{and } x^2 - 1 < 1; x^2 \geq 2$$

13. least value of $x^2 + x + 1 = \frac{3}{4}$ at $x = \frac{1}{2}$

$$f(-1) = 2$$

$$f(1) = 0 \text{ in case of decreasing function}$$

$$-a + b = 2, a + b = 0$$

$$\Rightarrow b = 1, a = -1$$

$$\Rightarrow f(x) = -x + 1$$

$$\therefore \text{least value of } \sqrt{x^2 + x + 1} \text{ is } \frac{\sqrt{3}}{2}$$

$$\text{Also } \sqrt{x^2 + x + 1} \leq 1$$

$$\text{Thus } \frac{\sqrt{3}}{2} \leq \sqrt{x^2 + x + 1} \leq 1$$

$$\Rightarrow \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) \leq \sin^{-1}(\sqrt{x^2 + x + 1}) \leq \sin^{-1}(1)$$

$$\Rightarrow \frac{\pi}{3} \leq \sin^{-1}(\sqrt{x^2 + x + 1}) \leq \frac{\pi}{2}$$

14. $f(x) = -px + q - rx; x \leq 0$

$$= f(x) = -px + q + rx; 0 < x \leq \frac{q}{p}$$

$$= f(x) = px - q + rx; \frac{q}{p} < x$$

$$\text{for } r = p; f^1(x) < 0; x \leq 0$$

$$= 0, \text{ if } 0 < x < \frac{q}{p}, > 0 \text{ if } x > \frac{q}{p}$$

from the graph(i) we have infinitely many points for min. value of $f(x)$

$$\text{for } r < p \quad f^1(x) < 0 \text{ if } x \leq 0$$

$$< 0 \text{ if } 0 < x \leq \frac{q}{p}$$

$$> 0 \text{ if } x > \frac{q}{p}$$

from the graph(ii), only one point of minimum $x = q/p$

$$\text{for } r > p \quad f^1(x) < 0; \text{ if } x \leq 0$$

$$> 0; \text{ if } x > 0$$

from the graph(iii), only one point of minimum

15. $f(x) = e^{x^2} + e^{-x^2} \rightarrow f^1(x)$

$$= 2x(e^{x^2} + e^{-x^2}) \geq 0 \forall x \in [0, 1]$$

$f(x)$ is a non-decreasing function on $[0, 1]$

$$\text{Hence max. value of } f(x) = f(1) = e + \frac{1}{e} = a$$

$$g^1(x) = (2x^2 + 1)e^{x^2} - 2x.e^{-x^2}$$

$$\geq 0 \forall x \in [0, 1] \rightarrow g(x)$$

is an increasing function $[0, 1]$

16. Let $y = f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2 + 1}}{x^2 + 1}\right)$

Clearly $y \in (0, \pi) \dots\dots(1)$

$$\text{Also } \cos y = \frac{\sqrt{2x^2 + 1}}{x^2 + 1} + 1 > 0$$

$$\cos y > 0$$

$$y < \frac{\pi}{2} \dots\dots\dots(2)$$

$$\text{from (1) and (2) } y \in \left[0, \frac{\pi}{2}\right) = R_f$$

FUNCTIONS

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17. $f(x)=1$ is false; $f(y) \neq 1$ is false and $f(z) \neq 2$ is true then Injection is possible

i.e $f(x) = 2$ (or) 3 ; $f(y) = 1$; $f(z) = 1$ (or) 3

18. $f(x) = 1 - x$; $0 \leq x \leq 2$

$$= x - 3; 2 < x \leq 4$$

$$g(x) = 2 + x; -1 \leq x \leq 0$$

$$= 2 - x; 0 < x \leq 3$$

$$(f \circ g)(x) = -1 - x; -2 \leq x \leq 0; -1 \leq x \leq 0$$

$$= x - 1; 0 < x \leq 2; -1 \leq x \leq 0$$

$$= x - 1; 0 \leq x \leq 2; 0 < x \leq 3$$

$$= -x - 1; -2 \leq x < 0; 0 < x \leq 3$$

19. Let $y = f(x) \rightarrow y = \frac{x^2 + 1}{x} \rightarrow x^2 - xy + 1 = 0$

$$\rightarrow x = \frac{y \pm \sqrt{y^2 - 4}}{2}$$

$$f^{-1} = \frac{y \pm \sqrt{y^2 - 4}}{2}$$

$$f^{-1}(x) = \frac{x \pm \sqrt{x^2 - 4}}{2}$$

$\therefore$ the range of $f^{-1}(x)$ is $[1, \infty)$, ignore the negative sign

$$\therefore f^{-1}(x) = \frac{x + \sqrt{x^2 - 4}}{2}$$

20. $f(x) + f(x+4)$

$$= f(x+2) + f(x+6) \forall x \in R$$

Replace x by $x+2$ and simplify

On solving we get $f(x) = f(x+8)$

21. $f(x) + f(x+3) = 0 \dots\dots(1)$

replace 'x' by 'x+3'

$$f(x+3) + f(x+6) = 0 \dots\dots(2)$$

$$(1) - (2) \rightarrow f(x) = f(x+6)$$

period of $f(x)$ is 6

$$g^1(x) = f(x+n) = f(x+(n-3)+3)$$

$$= f(x+(n-3)) (\because \text{period of 'f' is 3})$$

$$= f(x+3) = f(x) \rightarrow g^1(x) = f(x)$$

$$\rightarrow g^1\left(\frac{7}{3}\right) = f\left(\frac{7}{3}\right) = f\left(\frac{-2}{3} + 3\right) = f\left(\frac{-2}{3}\right) = 7$$

22. $x^2 f(x) - 2 \cdot f\left(\frac{1}{x}\right) = g(x) \Rightarrow$ replacing 'x' by

$$\frac{1}{x}, \text{ then } \frac{1}{x^2} \cdot f\left(\frac{1}{x}\right) - 2f(x) = g\left(\frac{1}{x}\right) \quad (\text{or})$$

$$2f\left(\frac{1}{x}\right) - 4x^2 f(x) = 2x^2 \cdot g\left(\frac{1}{x}\right) \dots\dots\dots(2)$$

$$(1) + (2) \Rightarrow f(x) = - \left(\frac{g(x) + 2x^2 \cdot g\left(\frac{1}{x}\right)}{3x^2} \right)$$

$$f(-x) = - \left(\frac{g(-x) + 2x^2 \cdot g\left(-\frac{1}{x}\right)}{3x^2} \right)$$

$$f(x) = -f(x)$$

($\therefore g(x)$ is an odd and $f(x)$ is an even function)

$$f(x) = 0 \Rightarrow f(5) = 0$$

23. $2f(xy) = (f(x))^y + (f(y))^x$ replacing 'y' by '1', then we get

$$2f(x) = f(x) + (f(1))^x$$

$$\Rightarrow f(x) = a^x$$

$$\text{Now } (a-1) \sum_{i=1}^n f(i) = (a-1)$$

$$\sum_{i=1}^n a^i = (a-1) [a + a^2 + a^3 + \dots\dots\dots + a^n]$$

$$= (a-1) \left(\frac{a(a^n - 1)}{a - 1} \right) = a^{n+1} - a$$

24. $f(xf(y)) = x^p y^q \Rightarrow [f(xf(y))]^{\frac{1}{p}} = x \cdot y^{\frac{q}{p}}$

$$x = \frac{[f(xf(y))]^{1/p}}{y^{q/p}} \dots\dots(1)$$

Now let $x \cdot f(y) = 1$

$$x = \frac{1}{f(y)} \dots\dots(2)$$

$$\text{Hence } f(x \cdot y^{q/p}) = x^p \cdot y^q$$

$$\text{put } y^{q/p} = z$$

$$f(xz) = (xz)^p$$

$$\Rightarrow f(\lambda) = \lambda^p \text{ -----(4)}$$

$$\text{from (3) and (4)} y^p = y^{q/p}$$

$$\Rightarrow p = \frac{q}{p} \Rightarrow p^2 = q$$

$$f(y) = \frac{y^{q/p}}{(f(1))^{1/p}} \Rightarrow f(1) = \frac{1}{(f(1))^{1/p}}$$

$$\Rightarrow f(1) = 1$$

$$\text{then } f(y) = y^{q/p} \text{ -----(3)}$$

$$25. \quad f(x+y) + f(x-y)$$

$$= 2f(x)f(y) \text{ -----(1)}$$

replace 'x' by 'y' and 'y' by 'x' in ---(1)

$$f(y+x) + f(y-x)$$

$$= 2f(y)f(x) \text{ -----(2)}$$

from (1) and (2), we get

$$f(y-x) = f(x-y) \text{ putting } y = 2x \text{ then}$$

$$f(x) = f(-x)$$

Hence $f(x)$ is an even function.

$$26. \quad \text{Replace 'x' by 'a' and 'y' by 'x-a'}$$

$$\Rightarrow f(a - (x-a)) = f(a) \cdot f(x-a) - f(0) \cdot f(x) \text{ -----(1)}$$

$$\text{put } x = 0; y = 0 \rightarrow f(0) = [f(0)]^2 - [f(a)]^2$$

$$\Rightarrow f(a) = 0, [\because f(0) = 1]$$

$$\text{from (1), } f(2a-x) = -f(x)$$

$$27. \quad 3f(x) + 2f\left(\frac{x+59}{x-1}\right) = 10x + 30$$

$$\text{put } x = 7$$

$$3f(7) + 2f(11) = 70 + 30 = 100$$

$$\text{put } x = 11$$

$$3f(11) + 2f(7) = 140 \rightarrow f(7) = 4$$

$$28. \quad f(x+a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2},$$

$$\text{put } x = x+a$$

$$f(x+a+a) = \frac{1}{2} + \sqrt{f(x+a) - (f(x+a))^2}$$

$$f(x+2a) = \frac{1}{2} + \sqrt{f(x+a)\{1 - f(x+a)\}}$$

$$= \frac{1}{2} + \sqrt{\left\{\frac{1}{2} + \sqrt{f(x) - (f(x))^2}\right\}\left\{\frac{1}{2} - \sqrt{f(x) - (f(x))^2}\right\}}$$

$$= \frac{1}{2} + \sqrt{\left(\frac{1}{2}\right)^2 + f(x) - (f(x))^2}$$

$$= \frac{1}{2} + f(x) - \frac{1}{2}$$

$$= f(x) \quad \because f(x) \geq 2$$

$$f(x+2a) = f(x)$$

period of 'f' is 2a

$$29. \quad f(x, y) = f(2x+2y, 2y-2x) \forall x,$$

$$y \in R \text{ -----(1)}$$

$$f(2(2x+2y)+2(2y-2x), 2(2y-2x)-2(2x+2y))$$

$$f(x, y) = [\text{from (1)}]$$

$$f(x, y) = f(8y-8x, 8x) \text{ -----(2)}$$

$$= f(8(-8x), -8(8y)) [\text{from (2)}]$$

$$f(x, y) = f(-64x, -64y) \text{ -----(3)}$$

$$= f(-64(-64x), -64(-64y)) \text{ from (3)}$$

$$f(x, y) = f(2^{12}x, 2^{12}y)$$

$$\text{replace 'y' by 0 } f(x, 0) = f(2^{12}x, 0)$$

$$\text{replace 'x' by } 2^y$$

$$f(2^y, 0) = f(2^{12} \cdot 2^y, 0)$$

$$= f(2^{12+y}, 0)$$

$$f(2^x, 0) = f(2^{12+x}, 0)$$

$$g(x) = g(12+x) \text{ for all 'x'}$$

period of 'g' is 12

$$30. \quad f(2) = \min(4, 2, \text{sgn}(17)) = \min(4, 2, 1) = 1$$

$$31. \quad f(x) = \frac{5}{4} \text{ on simplification}$$

Now $g(f(x)) = g\left(\frac{5}{4}\right) = 1 \Rightarrow y = 1$ is a straight line

$$32. \quad \text{In the sum 1 occurs } n \text{ times, } \frac{1}{2} \text{ occurs } (n-1) \text{ times, } \frac{1}{3} \text{ occurs } (n-2) \text{ times and so on}$$

FUNCTIONS

$$\begin{aligned}
 & \therefore f_1(1) + f_2(2) + f_3(3) + \dots + f_1(n) \\
 &= n(1) + (n-1)\frac{1}{2} + (n-2)\frac{1}{3} + \dots + 1 \cdot \frac{1}{n} \\
 &= n \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right) \\
 &= n \left(\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \dots + \frac{n-1}{n} \right) \\
 &= n \cdot f_1(n) - \left[\left(1 - \frac{1}{2} \right) + \left(1 - \frac{1}{3} \right) + \left(1 - \frac{1}{4} \right) + \dots + \left(1 - \frac{1}{n} \right) \right] \\
 &= n \cdot f_1(n) - [n - f_1(n)] = (n+1)f_1(n) - n
 \end{aligned}$$

MULTIPLE ANSWER QUESTIONS

33. Even extension is $f(-x)$
Odd extension is $-f(-x)$
34. $(f \circ g)(x) = \sqrt{3|\sin x| - \sin x - 2}$
 $\Rightarrow 3|\sin x| - \sin x - 2 \geq 0$
 If $\sin x > 0 \Rightarrow \sin x \geq 1$
 $\sin x = 1 \Rightarrow x \in 2n\pi + \frac{\pi}{2}$
 $\sin x < 0 \Rightarrow -1 \leq \sin x \leq \frac{1}{2}$
 $x \in \left[2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6} \right]$
35. $||x| - 1| - 5 > 0$
 $\Rightarrow |x| - 1 \geq 6$ (or) $|x| - 1 < -5$
 $\Rightarrow x \in (-\infty, -7] \cup [7, \infty)$
36. $2f(\sin x) + f(\cos x) = x$ ----(1)
 replace 'x' by ' $\frac{\pi}{2} - x$ ', we get
 $2f(\cos x) + f(\sin x) = \frac{\pi}{2} - x$ ----(2)
 from (1) and (2)
 $f(\sin x) = x - \frac{\pi}{6}$
 let $\sin x = y \Rightarrow x = \sin^{-1} y$
 $f(y) = \sin^{-1} y - \pi/6$
 Domain = $[-1, 1]$

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- $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$
 $-\frac{2\pi}{3} \leq \sin^{-1} x - \frac{\pi}{6} \leq \frac{\pi}{3}$
37. $(f \circ g)(x) = \sin(\ln|x|) \Rightarrow R_1 = [-1, 1]$
 $(g \circ f)(x) = \ln|\sin x| \Rightarrow R_2 = (-\infty, 0]$
38. Let $q(x) = K$ where $K \in R$ and
 $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots - a_nx^n = k$
 $a_0 = a_2 = a_3 = \dots = a_n = 0$ and $a_1 = 1$
 $p(x)$ ---> odd and of odd degree
39. $f(x) = 1 + \sin x; 0 \leq x \leq \frac{3\pi}{4}$
 $= 1 - \cos x; \frac{3\pi}{4} \leq x \leq \frac{3\pi}{2}$
 $= 1; \frac{3\pi}{2} \leq x \leq 2\pi$
 $g(x) = 1 - x; x \leq 0$
 $= 1; 0 \leq x \leq 2$
 $= x - 1; x \geq 2$
 $f(0) = 1 \Rightarrow g(f(1)) = 1$
 $f(1) = 1 + \sin 1$
 and $f(g(0)) = 1 + \sin 1$
40. If $f(x)$ is periodic
 if $4 - a^2$
 is a perfect square i.e. $\sqrt{4 - a^2}$ is a rational number.
 $\Rightarrow a = 0, 2, -2$
41. $f(x) = f^1(x) \cdot f^{11}(x)$
 degree of $f(x)$ is 3
 $n = n - 1 + n - 2 \dots > n = 3$
 $f(x)$ is an onto function
42. Clearly
 $f(1+x) = f(1-x) \dots >$ symmetric about the line $x = 1$
 max. value of $\ln(2x - x^2)$ as well as that of $\sin\left(\frac{\pi x}{2}\right)$ occurs at $x = 1$

$$\begin{aligned} \text{Domain} &\Rightarrow 2x - x^2 > 0 \Rightarrow x^2 - 2x < 0 \\ x &\in (0, 2) \text{ as } x \rightarrow 0 + (\text{or}) x \rightarrow 2 - \\ f(x) &\rightarrow -\infty \end{aligned}$$

PARAGRAPH QUESTIONS

$$43. \quad H(x) = \frac{f(x)}{g(x)} = \frac{1 - 2\sin^2 x}{\cos 2x} = 1 \text{ but } \cos(2x) \neq 0$$

$$2x \neq n\pi + \frac{\pi}{2}$$

$$\text{Domain} = R - \{(2n+1)\frac{\pi}{4}\}$$

$$\text{and Range} = 1$$

$$44. \quad F(x) = f(x) + g(x) =$$

$$1 - 2\sin^2 x + \cos 2x = 2\cos(2x)$$

$$-1 \leq \cos(2x) \leq 1 \Rightarrow -2 \leq 2\cos(2x) \leq 2$$

$$\Rightarrow \text{onto function}$$

$$45. \quad f(x) = 1 - 2\sin^2 x \text{ with period } \pi$$

$$g(x) = \cos(2x) \text{ with period } \pi$$

$$F(x) = 2\cos 2x \text{ with period } \pi \Rightarrow \pi + \pi + \pi = 3\pi$$

$$46. \quad G(x) = f(x) - g(x) =$$

$$1 - 2\sin^2 x - \cos(2x) = 0$$

$$H(x) = 1, \text{ union of their Domains is } R$$

$$47. \quad F(x) - G(x) \rightarrow 2g(x) = 0$$

$$\rightarrow \cos(2x) = 0 \rightarrow x$$

$$= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \dots, \frac{19\pi}{4}$$

$$\text{No. of solutions} = 10 \text{ and their sum} = 25\pi$$

$$48. \quad \text{As } f(296) = f(926) \rightarrow \text{many-one}$$

Further, the number whose digits increase from left to right

eg: 159 have no pre-image $\rightarrow$ into

$$49. \quad \text{Here } f(n) \text{ and } n \text{ leave the same remainder, when divided by 9. The remainder, when a +ve integer is divided by 9 is same as the sum of the digits of the number (until the sum becomes a single digit number)}$$

$$50. \quad \text{Clear from the definition}$$

$$51. \quad \text{If the degree is odd, then } f(-\infty) = -\infty; f(\infty) = \infty$$

$$52. \quad \text{Let } y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$$

$$\Rightarrow (y-1)x^2 + (4y-2)x + (3(y-2)) = 0.$$

$$\text{Here } \Delta \geq 0$$

$$(4-3c)y^2 - 4(1-c)y + (1-c) \geq 0$$

$$4-3c > 0 \text{ and}$$

$$16(1-c)^2 - 4(4-3c)(1-c) \leq 0$$

$$0 < c < 1$$

$$53. \quad \text{conceptual}$$

$$54. \quad f(x) - 2f\left(\frac{x}{2}\right) + f\left(\frac{x}{4}\right) = x^2$$

$$\text{Replace } x \text{ by } \frac{x}{2}$$

$$f\left(\frac{x}{2}\right) - 2f\left(\frac{x}{4}\right) + f\left(\frac{x}{8}\right) = \left(\frac{x}{2}\right)^2$$

$$f\left(\frac{x}{4}\right) - 2f\left(\frac{x}{8}\right) + f\left(\frac{x}{16}\right) = \left(\frac{x}{4}\right)^2$$

$$f\left(\frac{x}{2^n}\right) - 2f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right) = \left(\frac{x}{2^n}\right)^2$$

Adding we get

$$f(x) - f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^{n+1}}\right) + f\left(\frac{x}{2^{n+2}}\right)$$

$$= x^2 \left(1 + \frac{1}{2^2} + \dots + \frac{1}{2^{2n}} \right)$$

$$\text{As } n \rightarrow \infty$$

$$f(x) - f\left(\frac{x}{2}\right) = \frac{4x^2}{3}$$

repeating we get

$$f(x) - f(0) = \frac{16x^2}{9} \Rightarrow f(3) = 16 + f(0)$$

$$55. \quad f(x) - f(0) - x = 0$$

$$\Rightarrow \frac{16x^2}{9} - x = 0 \Rightarrow x = 0, \frac{9}{16}$$

$$56. \quad f^1(x) = \frac{32x}{9} \Rightarrow f^1(0) = 0$$

$$57, 58, 59.$$

$$(f(x))^2 \cdot f\left(\frac{1-x}{1+x}\right) = 64x \dots (1)$$

$$\text{put } \frac{1-x}{1+x} = y \text{ and } x = \frac{1-y}{1+y} \text{ we get}$$

$$\left\{ f\left(\frac{1-y}{1+y}\right) \right\}^2 f(y) = 64 \left(\frac{1-y}{1+y} \right)$$

$$\Rightarrow f(x) \cdot \left\{ f\left(\frac{1-x}{1+x}\right) \right\}^2 = 64 \left(\frac{1-x}{1+x}\right) \text{-----} (2)$$

from (1)² and (2), we have

$$\frac{(f(x))^4 \left\{ f\left(\frac{1-x}{1+x}\right) \right\}^2}{f(x) \left\{ f\left(\frac{1-x}{1+x}\right) \right\}^2} = \frac{(64x)^2}{64 \left(\frac{1-x}{1+x}\right)}$$

$$\Rightarrow (f(x))^3 = 64x^2 \left(\frac{1+x}{1-x}\right)$$

$$\Rightarrow f(x) = 4 \cdot x^{2/3} \left(\frac{1+x}{1-x}\right)^{1/3}$$

Now solve other two questions

60-62

$$f(x) = \begin{cases} x+1, & x \leq 1 \\ 2x+1, & 1 < x \leq 2 \end{cases}$$

$$g(x) = \begin{cases} x^2, & 1 \leq x < 2 \\ x+2, & 2 \leq x < 3 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} g(x)+1, & g(x) \leq 1 \\ 2g(x)+1, & 1 < g(x) \leq 2 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} x^2+1, & x^2 \leq 1, & -1 \leq x < 2 \\ x+2+1, & x+2 \leq 1, & 2 \leq x \leq 3 \\ 2x^2+1, & 1 < x^2 \leq 2, & -1 \leq x < 2 \\ 2(x+2)+1, & 1 < x+2 \leq 2, & 2 \leq x \leq 3 \end{cases}$$

$$\Rightarrow f(g(x)) = \begin{cases} x^2+1, & -1 \leq x \leq 1 \\ 2x^2+1, & 1 < x \leq \sqrt{2} \end{cases}$$

60. Hence, the domain of $f(x)$ is $[-1, \sqrt{2}]$

61. For $-1 \leq x \leq 1$ we have

$$x^2 \in [0, 1] \Rightarrow x^2 + 1 \in [1, 2]$$

For $1 < x \leq \sqrt{2}$, we have

$$x^2 \in (1, 2] \Rightarrow 2x^2 + 1 \in (3, 5]$$

Hence, the range is $[1, 2] \cup (3, 5]$

62. For $f(g(x)) = 2 \Rightarrow x^2 + 1 = 2$ and

$$2x^2 + 1 = 2 \Rightarrow x = \pm 1 \text{ or } x = \pm \frac{1}{\sqrt{2}}$$

$\Rightarrow x = \pm 1$ only. **Hence, 2 roots**

63. $f'(x) = 0 \Rightarrow \tan x = x$

Draw the graphs of

$$y = \tan x \text{ \& } y = x$$

64. for $x \in \left(0, \frac{\pi}{2}\right), f(x) > 0$

as $\tan x > x$

$$\begin{aligned} \text{65. Reqd Area} &= \int_0^\alpha f(x) dx - \int_\alpha^{2\pi} f(x) dx \\ &= -[2 \cos x + x \sin x]_0^\alpha + [2 \cos x + x \sin x]_\alpha^{2\pi} \\ &= 4 - 2(2 + \alpha^2) \cos \alpha \end{aligned}$$

MATRIX-MATCHING QUESTIONS

66. a --- > Let $y = f(x) = \cot^{-1}(2x - x^2 - 2)$

clearly $y \in (0, \pi)$ -----(1)

and

$$\cot y = 2x - x^2 - 2 = -1 - (x-1)^2 \leq 1$$

$$\cot y \leq -1$$

$$y \in \left(\frac{3\pi}{4}, \pi\right) \text{ --- > into}$$

more over $f^1(x)$ changes its sign in the neighbourhood of $x=2$
many-one

$$b \text{ --- > } -e^{ax} \leq e^{ax} \sin(bx) \leq e^{ax} \text{ --- >}$$

many-one, onto

$$c \text{ --- > } f^1(x) = x \text{ --- > } f^1(x) > 0 \text{ --- >}$$

many-one i.e., $f(x)$ is an increasing function

$$\text{range} = (f(0), f(\infty))$$

$$= (2, \infty) \neq \text{co-domain} \text{ --- > into}$$

d --- > clearly identity function --- > one-one and onto

67. $a \rightarrow (x^2 - 1) \cos x \geq 0 \rightarrow x^2 - 1 \geq 0$ and

$$\cos x \geq 0 \text{ (or) } x^2 - 1 \leq 0 \text{ and } \cos x \leq 0$$

$$-\frac{\pi}{2} \leq x \leq -1 \text{ (OR) } 1 \leq x \leq \frac{\pi}{2}$$

$$b \rightarrow [x]^2 = x + 2\{x\} \rightarrow \{x\} = \frac{[x]^2 - [x]}{3}$$

$$\text{Now } 0 \leq \frac{[x]^2 - [x]}{3} < 1$$

$$\rightarrow [x] \in \left[b \rightarrow [x]^2 = x + 2\{x\} \rightarrow \{x\} = \frac{[x]^2 - [x]}{3} \right]$$

$$\rightarrow [x] = -1, 0, 1, 2$$

$$\{x\} = \frac{2}{3}, 0, 0, \frac{2}{3} \rightarrow x = \frac{-1}{3}, 0, 1, \frac{8}{3}$$

$$c \rightarrow -1 \leq x \leq 1 \rightarrow -\frac{\pi}{4} \leq \tan^{-1} x \leq \frac{\pi}{4}$$

$$\therefore f(x) = \frac{\pi}{2} + \tan^{-1} x \rightarrow \frac{\pi}{4} \leq f(x) \leq \frac{3\pi}{4}$$

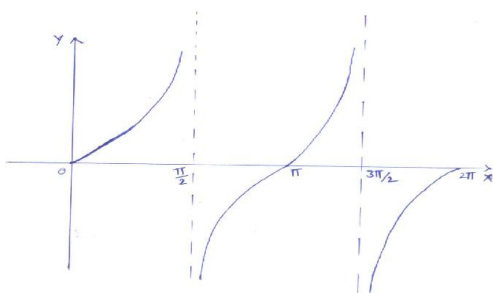
$$\rightarrow [f(x)] = 0, 1, \text{ or } 2$$

$$d \rightarrow \ln(\cos(\sin x)) \geq 0$$

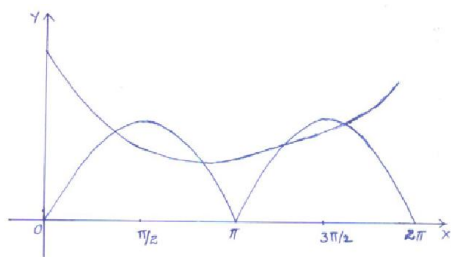
$$\rightarrow \cos(\sin x) \geq 1 \rightarrow \cos(\sin x) = 1$$

$$\rightarrow \ln(\cos(\sin x)) = 0$$

68. $a \rightarrow$ draw the graphs of $y = \tan x$ & $y = \frac{1}{x^2}$
from the graph, it is clear that it will have two real roots



- $b \rightarrow$ draw the graphs $y = 2^{\cos x}$ and $y = |\sin x|$
no. of P.O.I = 4

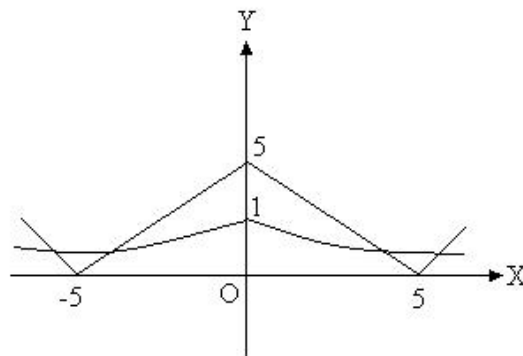


$c \rightarrow \therefore f(|x|) = 0$ has 8 real roots

$\rightarrow f(x) = 0$ has 4 real roots $\therefore f(x)$ is a polynomial of degree 5, $f(x)$ cannot have even no. of real roots

$f(x)$ has all the five roots real and one root is

$f(x)$ has all the five roots real and one root is negative



- $d \rightarrow$ Draw the graphs of $y = 7^{-|x|}$ and $y = |5 - |x||$

69. $a \rightarrow$ when $-1 \leq x < 0$ $f(x) \in [0, 1]$ and $f(x) = x + 1$

$$\therefore f(f(x)) = (x+1)^2 - 1 = x^2 + 2x$$

$$b \rightarrow \frac{[\cos(2x) + 1][\sec^2 x + 2 \tan x]}{2}$$

$$= \frac{1 + \tan^2 x + 2 \tan x}{1 + \tan^2 x}$$

$$\text{i.e. } f\left(\frac{2 \tan x}{1 + \tan^2 x}\right) = 1 + \frac{2 \tan x}{1 + \tan^2 x}$$

$$\therefore f(x) = 1 + x$$

$$c \rightarrow \text{put } x = y = 0$$

$$\rightarrow f(1) = (1+1)^2 = 2^2;$$

$$\text{put } x = 0; y = 1 \rightarrow f(2) = 3^2$$

$$d \rightarrow \text{when } 4 < x < 5 \rightarrow f(x)$$

$$= 2x + 3 \rightarrow f^{-1}(x) = \frac{x-3}{2}$$

$$f(\theta) = \frac{4 \sin^2 \theta \cos \theta - \cos \theta + \sin \theta}{4(\cos \theta + \sin \theta)}$$

$$= \frac{1}{2} \sin \theta (\sin \theta + \cos \theta) - \frac{1}{4}$$

$$= \frac{1}{4} [\sin(2\theta) - \cos(2\theta)] \in \left[\frac{-1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right]$$

FUNCTIONS

ASSERTION - REASON QUESTIONS

70. $f(x) = g(x) \rightarrow L.H.S = 1 + e^{\cot^2 x} \geq 2$
 As $\sqrt{2}|\sin x| - 1 \leq \text{and}$

$$\frac{1 - \cos(2x)}{1 + \sin^4 x} = \frac{2\sin^2 x}{1 + \sin^4 x}$$

$$= \frac{2}{\frac{1}{\sin^2 x} + \sin^2 x} \leq 1$$

 $\therefore R.H.S \leq 2 \therefore \text{Equality holds good}$
 $L.H.S = R.H.S = 2$
 Which is possible when
 $\cot^2 x = 0 \text{ and } |\sin x| = 1$
 $\rightarrow x = (2n+1)\frac{\pi}{2}; n \in I$
 $\therefore S_1 \text{ is correct}$
 Clearly $\therefore S_2$ need not be correct always
 $\therefore S_1 \text{ is true and } S_2 \text{ is false}$
71. Conceptual
 72. Conceptual
 73. $f(-1) = 17$ and $f(1) = -13$
 clearly S_1 is also true

INTEGER QUESTIONS

74. $f(x) + f(x+3) = 0 \dots (1)$
 replace 'x' by 'x+3'
 $f(x+3) + f(x+6) = 0 \dots (2)$
 $(1) - (2) \Rightarrow f(x) = f(x+6)$
 $\Rightarrow \text{period of } f(x) \text{ is } 6$
75. Period of $\cos(\sin(nx))$ is $\frac{\pi}{n}$ and the period of
 $\tan\left(\frac{x}{n}\right)$ is πn
 Now L.C.M of $\frac{\pi}{n}, \pi n$ is $\pi n \Rightarrow n = 6$
76. $f(x+2) = f(x+1+1) = f(x+1) = f(x)$
 Period = 2

JEE ADVANCED - VOL - III

77. $|f(x) + g(x)| \leq |f(x)| + |g(x)| \dots (1)$
 but given that
 $|f(x) + f(x)| \geq |f(x)| + |g(x)| \dots (2)$
 from (1) and (2)
 $|f(x) + g(x)| = |f(x)| + |g(x)|$
 this is possible only when
 $f(x) \cdot g(x) \geq 0$ but given that
 $f(x) \cdot g(x) \leq 0$
 $\rightarrow f(x)g(x) = 0 \rightarrow f(x) = 0 (\because g(x) \neq 0)$
 $\therefore \sum_{r=1}^{100} f(r) = 0$
78. Conceptual
 79. $b^2 - 4ac < 0$
 80. Properties of G.I.F.
 81.
$$f(x) = \frac{1 + x^4 + x + x^5 + x^2 + x^6}{x^3}$$

$$= \left(\frac{1}{x^3} + x^3\right) + \left(x + \frac{1}{x}\right) + \left(x^2 + \frac{1}{x^2}\right)$$

$$\geq 2\sqrt{\frac{1}{x^3} \cdot x^3} + 2\sqrt{x \cdot \frac{1}{x}} + 2\sqrt{x^2 \cdot \frac{1}{x^2}} = 6$$

 $\left(\text{use } a + \frac{1}{a} \geq 2, a > 0\right)$
 $\Rightarrow f(x) \geq 6$
 $\therefore R_f = [6, \infty)$
82. $\because x \in (\sqrt{2}, \sqrt{3})$
 $\Rightarrow \sqrt{2} < x < \sqrt{3} \quad \text{or} \quad \frac{1}{\sqrt{3}} < \frac{1}{x} < \frac{1}{\sqrt{2}}$
 $\therefore \left\{\frac{1}{x}\right\} = \frac{1}{x} \text{ and } 2 < x^2 < 3$
 or $0 < x^2 - 2 < 1$
 $\therefore \{x^2\} = x^2 - 2$
 Given $\left\{\frac{1}{x}\right\} = \{x^2\} \Rightarrow \frac{1}{x} = x^2 - 2 \dots (i)$
 $\Rightarrow x^3 - 2x - 1 = 0 \Rightarrow (x+1)(x^2 - x - 1) = 0$
 $x \neq 1$

$$\begin{aligned}
 \therefore x^2 - x - 1 &= 0 \Rightarrow x^2 = x + 1 \\
 \Rightarrow x^4 &= x^2 + 2x + 1 = x + 1 + 2x + 1 \\
 \Rightarrow x^4 &= 3x + 2 \quad \dots\dots(ii) \\
 \therefore x^4 &= 3x + 2 \\
 \Rightarrow x^4 - \frac{3}{x} &= (3x + 2) - 3(x^2 - 2) \\
 &\text{(from eqs. (i) and (ii))} \\
 &= (3x + 2) - 3(x + 1 - 2) \\
 &= 5. \quad (\because x^2 = x + 1)
 \end{aligned}$$

83. Given $f(x) \neq 0$

$\therefore f(x)$ is symmetrical about the line $y = x$, then

$$\begin{aligned}
 f^{-1}(x) &= f(x) = y = x \\
 \Rightarrow f^2(x) &= (f^{-1}(x))^2 - \lambda x f(x) f^{-1}(x) \\
 &\quad + 3x^2 f(x)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow x^2 &= x^2 - \lambda x \cdot x \cdot x + 3x^2 \cdot x \\
 \Rightarrow \lambda x^3 &= 3x^3 \cdot \\
 \therefore \lambda &= 3, x \neq 0.
 \end{aligned}$$

84. $\therefore x^2 + 4y^2 = 4$

$$\Rightarrow \left(\frac{x}{2}\right)^2 + \left(\frac{y}{1}\right)^2 = 1.$$

Let $x = 2 \cos \theta, y = \sin \theta$

$$\begin{aligned}
 \Rightarrow f(\theta) &= (2 \cos \theta)^2 + (\sin \theta)^2 - (2 \cos \theta)(\sin \theta) \\
 &= 4 \cos^2 \theta + \sin^2 \theta - \sin 2\theta \\
 &= 3 \cos^2 \theta + (\cos^2 \theta + \sin^2 \theta) - \sin \theta \\
 &= \frac{3}{2}(1 + \cos 2\theta) + 1 - \sin 2\theta = \frac{3}{2} \cos 2\theta - \sin 2\theta + \frac{5}{2} \\
 &\Rightarrow \frac{5}{2} - \sqrt{\left(\frac{3}{2}\right)^2 + (-1)^2} \leq \frac{3}{2} \cos 2\theta - \sin 2\theta \\
 &\quad \frac{5}{2} \leq \frac{5}{2} + \sqrt{\left(\frac{3}{2}\right)^2 + (-1)^2} \\
 &\Rightarrow \frac{5 - \sqrt{13}}{2} \leq f(\theta) \leq \frac{5 + \sqrt{13}}{2}.
 \end{aligned}$$

$$\text{Hence, } L = \frac{5 - \sqrt{13}}{2} \text{ and } M = \frac{5 + \sqrt{13}}{2}.$$

$$\begin{aligned}
 \therefore LM &= \left(\frac{5 - \sqrt{13}}{2}\right) \left(\frac{5 + \sqrt{13}}{2}\right) = \frac{25 - 13}{4} \\
 &= \frac{12}{4} = 3.
 \end{aligned}$$

85: We have

$$f(x+1) + f(x-1) = \sqrt{3} f(x) \quad \forall x \in R \dots(i)$$

Replacing x by $x-1$ and $x+1$ in (i), then

$$f(x) + f(x-2) = \sqrt{3} f(x-1) \dots(ii)$$

$$\text{and } f(x+2) + f(x) = \sqrt{3} f(x+1) \dots(iii)$$

Adding (ii) and (iii), we get

$$\begin{aligned}
 2f(x) + f(x-2) + f(x+2) \\
 = \sqrt{3}(f(x-1) + f(x+1))
 \end{aligned}$$

$$\Rightarrow 2f(x) + f(x-2) + f(x+2) = \sqrt{3} \cdot \sqrt{3} f(x)$$

[From (i)]

$$\therefore f(x+2) + f(x-2) = f(x) \dots(iv)$$

Replacing x by $x+2$ in equation (iv),

$$\text{then } f(x+4) + f(x) = f(x+2) \dots(v)$$

Adding equations (iv) and (v), we get

$$f(x+4) + f(x-2) = 0 \dots(vi)$$

Again replacing x by $x+6$ in (vi), then

$$f(x+10) + f(x+4) = 0 \dots(vii)$$

Subtracting (vi) from (vii), we get

$$f(x+10) - f(x-2) = 0 \dots(viii)$$

Replacing x by $x+2$ in (viii), then

$$f(x+12) - f(x) = 0$$

$$\text{or } f(x+12) = f(x)$$

Hence $f(x)$ is periodic function with period 12.